ABSTRACT

The rapid advancement in robotics and automation has created an ever-growing demand for low-cost, intelligent, and flexible robotic systems capable of performing complex tasks in dynamic environments. Conventional robotic systems are typically fixed in position and lack the mobility required for versatile deployment in industrial and educational settings. To address these limitations, this project presents the design, construction, and comprehensive evaluation of a mobile robotic system that integrates a BCN3D Moveo open-source 5 Degrees of Freedom (DOF) robotic arm mounted on a 3-wheeled mecanum omnidirectional mobile base.

The robotic arm assembly consists of five stepper-motor-driven joints providing full spatial dexterity and one servo-controlled gripper, controlled by an Arduino Nano microcontroller interfaced with A4988 stepper motor driver modules via a CNC shield. The mobile platform employs three mecanum wheels driven by Johnson DC motors through L298N motor driver modules, governed by a NodeMCU (ESP8266/ESP32) microcontroller. The use of mecanum wheels enables omnidirectional locomotion — forward, backward, lateral, and diagonal movement — without requiring the robot to change its orientation.

Wireless communication between the operator's handheld joystick controller and the robot is implemented using the ESP-NOW protocol, a connectionless, peer-to-peer communication standard developed by Espressif Systems. This protocol offers low-latency (5–20 ms), reliable data transmission without dependence on Wi-Fi infrastructure or the internet, making it particularly suitable for real-time embedded control applications.

A motion recording and playback system has also been implemented, enabling the robot to memorize sequences of joint positions and wheel movements and replay them autonomously. This feature significantly enhances the system's utility for repetitive task automation in laboratory and educational environments. The entire system has been realized at an approximate cost of Rs. 5,000, demonstrating the feasibility of developing capable robotic platforms on a budget.

The project successfully demonstrates the seamless integration of embedded systems, stepper motor control, wireless communication, and robotic kinematics into a unified, functional platform. Experimental evaluations confirm low-latency communication, smooth omnidirectional mobility, and approximately 95% motion playback accuracy. The

system offers a versatile and cost-effective solution with wide applicability in educational robotics, industrial automation research, and prototyping environments.

Keywords: *Robotic Arm, BCN3D Moveo, Mecanum Wheels, ESP-NOW, Arduino, NodeMCU, Motion Recording, Stepper Motor, Omnidirectional Robot, Embedded Systems*

LIST OF ABBREVIATIONS

Abbreviation	Full Form
DOF	Degrees of Freedom
ESP-NOW	Espressif Systems - No Wi-Fi Protocol
CNC	Computer Numerical Control
MCU	Microcontroller Unit
PWM	Pulse Width Modulation
PCB	Printed Circuit Board
UART	Universal Asynchronous Receiver-Transmitter
SPI	Serial Peripheral Interface
I2C	Inter-Integrated Circuit
ROS	Robot Operating System
AI	Artificial Intelligence
ECE	Electronics and Communication Engineering
DC	Direct Current
AC	Alternating Current
GPIO	General Purpose Input/Output
SDK	Software Development Kit
API	Application Programming Interface
MAC	Media Access Control

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CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Robotics has emerged as one of the most transformative fields in modern engineering, fundamentally reshaping industries ranging from manufacturing and logistics to healthcare and defense. Early robotic systems were characterized by fixed, stationary platforms designed to perform highly specific, repetitive tasks within a confined workspace. While these systems significantly improved productivity and precision in industrial environments such as automotive assembly lines, their inherent immobility severely limited their applicability to a broader range of real-world tasks.

The evolution of robotics has progressively moved toward systems that combine manipulation capability with mobility — the so-called mobile manipulators. Such systems are capable of navigating through dynamic environments, reaching objects at various locations, and performing dexterous manipulation tasks. This convergence of mobility and manipulation opens up entirely new domains of application, including hospital logistics, warehouse automation, disaster response, and educational research.

Simultaneously, the open-source hardware movement has democratized access to advanced robotic platforms. Projects such as the BCN3D Moveo, originally developed in Spain, have made sophisticated multi-DOF robotic arms accessible to students, researchers, and hobbyists worldwide, drastically reducing the cost barrier to entry. The availability of open-source designs, combined with low-cost microcontrollers such as the Arduino and NodeMCU families, has made it feasible to build functional robotic systems at a fraction of the cost of commercial alternatives.

This project is motivated by the desire to bring together these developments — open-source robotic arm design, omnidirectional mobile base technology, and low-power wireless communication — into a single cohesive and capable robotic system that is affordable, extensible, and suitable for academic exploration.

1.2 Problem Statement

Conventional robotic arms, while precise and capable, suffer from a fundamental limitation: they are anchored to a fixed base and operate within a limited reach

envelope. This constraint prevents them from being deployed in environments where the robot must move between workstations, navigate around obstacles, or interact with objects distributed across a wide area.

Moreover, traditional wired control systems impose restrictions on operational range and maneuverability. The absence of a motion recording feature in basic control systems also means that operators must manually guide the robot through every operation, making repetitive tasks tedious and error-prone. There exists a clear need for an integrated robotic platform that addresses the following challenges:

- Lack of mobility in existing robotic arm systems, limiting their deployment flexibility.
- Absence of omnidirectional motion in conventional wheeled robots, which can only move forward and backward or rotate.
- Dependence on wired control or internet-based wireless protocols that introduce latency and infrastructure requirements.
- Absence of motion automation capabilities for repeating complex multi-joint sequences.
- High cost of commercial mobile manipulators that places them out of reach for educational institutions.

1.3 Objectives of the Project

The primary objectives of this project are as follows:

1. To design and fabricate a 5 DOF robotic arm based on the BCN3D Moveo open-source platform, utilizing stepper motors for precise angular positioning of each joint.
2. To develop a 3-wheel mecanum omnidirectional mobile base capable of movement in any direction without changing the robot's heading orientation.
3. To implement wireless communication between a joystick controller and the robot using the ESP-NOW protocol for low-latency, infrastructure-free real-time control.
4. To develop a motion recording and playback system that enables the robot to store and autonomously replay sequences of arm and base movements.
5. To achieve the above objectives within a cost constraint of approximately Rs. 5,000 to demonstrate affordability.

1.4 Scope of the Project

This project encompasses the mechanical assembly, electronic hardware integration, embedded firmware development, and functional testing of the robotic system. The scope includes the design of the power distribution system, the selection and

interfacing of sensors and actuators, the implementation of the wireless communication stack, and the development of the motion automation algorithm.

The scope of this project does not include the development of computer vision or AI-based autonomous object detection and grasping systems, real-time trajectory planning using inverse kinematics solvers, or integration with the Robot Operating System (ROS) — though these are identified as important directions for future work. The project focuses deliberately on core embedded systems and robotics fundamentals to serve as a solid pedagogical platform.

1.5 Report Organization

This report is organized into six chapters. Chapter 2 presents a detailed literature survey covering relevant work in open-source robotic arms, mecanum wheel kinematics, ESP-NOW wireless communication, and motion recording systems. Chapter 3 describes the system design, including hardware components, circuit design, and mechanical integration. Chapter 4 covers the software implementation, detailing the control algorithms, communication protocol, and motion recording logic. Chapter 5 presents the experimental results and performance evaluation. Chapter 6 concludes the report with a summary of achievements and directions for future work. References and appendices are provided at the end.

CHAPTER 2: LITERATURE SURVEY

2.1 Overview

This chapter presents a review of existing literature and technical documentation relevant to the key subsystems of this project: open-source robotic arm platforms, mecanum wheel omnidirectional bases, ESP-NOW wireless communication, and motion recording systems. The survey is intended to identify the state of the art, establish the theoretical foundations, and justify the design choices made in this project.

2.2 Open-Source Robotic Arms

The BCN3D Moveo is an open-source 5 DOF robotic arm designed and released by BCN3D Technologies, a spin-off of the Escola d'Enginyeria de Barcelona Est (EEBE), part of the Polytechnic University of Catalonia in Spain. It was designed as a low-cost alternative to industrial robotic arms, with all structural parts intended to be 3D-printed in PLA or ABS plastic, making it highly accessible and customizable.

The Moveo arm features five rotational joints driven by NEMA 17 or NEMA 23 stepper motors, with a sixth servo motor for gripper actuation. Each joint's motor is connected via a toothed belt and pulley reduction system, which amplifies the torque output while reducing the angular resolution requirement on the motor. This design is widely documented on GitHub and has inspired numerous academic and hobbyist projects globally.

Prior research has validated the Moveo arm as a reliable educational robotics platform. Studies have explored its use in pick-and-place operations, drawing machines, and laboratory automation. The open-source nature of the project allows modifications to joint geometry, payload capacity, and control electronics to be made freely, which is leveraged extensively in this work.

2.3 Mecanum Wheel Omnidirectional Systems

Mecanum wheels, invented by Swedish engineer Bengt Ilon in 1972 and originally patented under the name 'Ilon wheel', are a type of omnidirectional wheel that enables a vehicle to move in any direction without changing its orientation. Each mecanum wheel is fitted with a set of barrel-shaped passive rollers arranged at 45 degrees to the wheel's rotational axis. The interaction between these rollers and the ground

surface produces a resultant force that can be decomposed into longitudinal and lateral components.

For a 3-wheeled mecanum configuration, the kinematics are derived from the constraint equations of each wheel. Unlike the standard 4-wheel configuration, the 3-wheel layout offers compactness but requires careful placement of wheels (typically at 120-degree intervals or an asymmetric configuration) to ensure full omnidirectionality. The velocity equations governing a 3-wheel mecanum robot are given by:

Eq. 2.1	$V1 = Vy + Vx + W(Lx + Ly)$
Eq. 2.2	$V2 = Vy - Vx - W(Lx + Ly)$
Eq. 2.3	$V3 = Vy - W(Lx + Ly)$

where V1, V2, V3 are the angular velocities of the three wheels, Vx and Vy are the desired translational velocity components, W is the desired rotational velocity, and Lx, Ly are the geometric distances from the robot's center of rotation to each wheel. Previous implementations of 3-wheel mecanum robots in research literature have demonstrated their effectiveness in confined spaces and have validated the kinematic model presented above.

2.4 ESP-NOW Wireless Communication Protocol

ESP-NOW is a connectionless Wi-Fi communication protocol developed and maintained by Espressif Systems for use with their ESP8266 and ESP32 microcontroller families. Unlike standard Wi-Fi communication, which requires both devices to be connected to a router or access point, ESP-NOW allows two or more ESP devices to communicate directly with each other using their Wi-Fi hardware, without any external network infrastructure.

Key technical characteristics of ESP-NOW include a maximum payload size of 250 bytes per packet, communication range of approximately 200 meters in open air, typical end-to-end latency of 5–20 milliseconds, support for both unicast and broadcast transmission modes, and the ability to pair up to 20 peer devices simultaneously. These properties make ESP-NOW highly suitable for real-time robotic control applications, where low latency and reliability are critical.

Published research on ESP-NOW has evaluated its performance for drone control, sensor networks, and multi-robot communication. Results consistently show that ESP-NOW outperforms standard Wi-Fi-based UDP communication in terms of latency and reliability for short to medium range applications, validating its selection for this project.

2.5 Motion Recording Systems in Robotics

Motion recording, also known as teach-by-demonstration or kinesthetic teaching, is a technique in which a robot records a sequence of states (joint positions, velocities, or tool paths) as an operator manually guides it through a task, and then replays these states to reproduce the demonstrated motion autonomously. This approach eliminates the need for complex task programming and makes robotic automation accessible to non-expert operators.

Commercial implementations of this concept include the 'lead-through programming' feature in industrial robots from manufacturers such as KUKA and ABB. In low-cost embedded robotics, similar functionality can be implemented by logging timestamped joint encoder readings into non-volatile memory (EEPROM or SD card) and replaying them in a timed sequence. Research has shown that motion recording accuracy depends primarily on the sampling rate, encoder resolution, and the mechanical repeatability of the actuation system.

2.6 Research Gap and Project Contribution

The reviewed literature reveals that while robotic arms and mecanum wheel bases have each been well-studied individually, their integration into a single wireless-controlled mobile manipulator with motion recording capability at a sub-Rs. 5,000 price point has not been extensively documented. This project addresses this gap by providing a practical, reproducible, and documented implementation of such a system using widely available open-source hardware and free embedded development tools.

CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY

3.1 Overall System Architecture

The proposed system is composed of three primary subsystems that work in concert: (1) the Joystick Controller Unit, which captures operator input and transmits commands wirelessly; (2) the Mobile Base Unit, consisting of the NodeMCU and the 3-wheel mecanum drive system; and (3) the Robotic Arm Unit, consisting of the Arduino Nano, the CNC shield with stepper drivers, and the five-joint Moveo arm assembly.

The Joystick Controller Unit houses a NodeMCU in transmitter mode equipped with two analog joysticks and push buttons. One joystick controls the mobile base (X and Y translational motion, and rotation), while the second joystick controls the arm's base rotation and shoulder joint. Additional push buttons are mapped to the remaining arm joints and the gripper servo. A dedicated button triggers motion recording and another triggers playback.

ESP-NOW packets are broadcast by the controller and received by a second NodeMCU on the robot (the receiver). The receiver NodeMCU processes the base motion commands directly and forwards arm commands to the Arduino Nano over a serial UART link. The Arduino interprets these commands and drives the stepper motors through the A4988 drivers accordingly.

3.2 Hardware Components and Specifications

Component	Specification / Description
Arduino Nano	ATmega328P MCU, 16 MHz clock, 32 KB Flash, 2 KB SRAM, 22 I/O pins
NodeMCU ESP8266	Tensilica L106, 80/160 MHz, 4 MB Flash, 802.11 b/g/n Wi-Fi, ESP-NOW support
NodeMCU ESP32	Dual-core Xtensa LX6, 240 MHz, 4 MB Flash, Wi-Fi + Bluetooth, ESP-NOW
A4988 Stepper Driver	VREF-adjustable, up to 2A/phase, microstepping 1/1 to 1/16, overcurrent protection

Component	Specification / Description
L298N Motor Driver	Dual H-bridge, 2A per channel, 5–46V supply, PWM speed control, direction control
NEMA 17 Stepper Motor	1.8° step angle, 42 mm frame, 1.5A/phase, 0.45 Nm holding torque
NEMA 23 Stepper Motor	1.8° step angle, 57 mm frame, 2.8A/phase, 1.26 Nm holding torque
Johnson DC Motor	12V, 100–300 RPM (with gearbox), 0.5–1.5 Nm rated torque
Servo Motor (MG996R)	180° range, 9.4 kgf·cm stall torque, 4.8–6V operating voltage
Arduino CNC Shield V3	Supports 4 A4988 driver slots, GRBL-compatible pin mapping, RESET/ABORT headers
Analog Joystick Module	PS2-type, dual-axis potentiometer, push-button select, 3.3/5V compatible
12V DC Power Supply	5A rated, regulated, with screw terminal output
7805 Voltage Regulator	5V fixed output, 1A, TO-220 package, used for logic-level power supply
Mecanum Wheels (3x)	100 mm diameter, 45° roller angle, plastic hub, rubber rollers

The total bill of materials for the project was assembled at an approximate cost of Rs. 5,000, with the major cost contributors being the stepper motors (~Rs. 1,200), the mecanum wheel set (~Rs. 800), the NodeMCU modules (~Rs. 400), and the 3D-printed or acrylic structural components (~Rs. 600). This validates the project's objective of demonstrating a cost-effective robotic platform.

3.3 Robotic Arm Design and Joint Configuration

The BCN3D Moveo arm replicates the kinematic structure of a 6-axis industrial robotic arm in a 5-DOF configuration. The five joints are designated as follows:

Joint	Motion Type	Motor	Range of Motion	Belt Ratio
Joint 1 (Base)	Rotation (Z-axis)	NEMA 23	0° – 360°	3:1
Joint 2 (Shoulder)	Elevation	NEMA 23	-30° – +90°	5:1
Joint 3 (Elbow)	Flexion/ Extension	NEMA 17	0° – 135°	4:1

Joint	Motion Type	Motor	Range of Motion	Belt Ratio
Joint 4 (Wrist Pitch)	Pitch	NEMA 17	-90° – +90°	2:1
Joint 5 (Wrist Roll)	Roll	NEMA 17	-180° – +180°	1:1
Gripper	Open/Close	MG996R Servo	0° – 90°	N/A

The structural components of the arm are fabricated from 3D-printed PLA with strategic reinforcements at high-stress joints. All rotating joints employ 625ZZ miniature ball bearings for smooth, low-friction rotation. The toothed belt and pulley transmission between each motor and its corresponding joint provides both torque amplification and mechanical advantage, enabling the arm to lift payloads of up to approximately 500 grams at full extension.

3.4 Mobile Base Design

The mobile base is a triangular frame constructed from aluminum extrusion profile and 3D-printed joint connectors. Three mecanum wheels are positioned symmetrically around the chassis, each driven by a Johnson DC motor through a gear reduction box. The 3-wheel mecanum configuration is chosen over the conventional 4-wheel layout to reduce the overall weight and mechanical complexity of the base, at the expense of slightly reduced load capacity.

Each motor is connected to one channel of an L298N dual H-bridge motor driver module. Two L298N modules are used in total, with the third motor connected to one channel of the second module. PWM signals from the NodeMCU's GPIO pins are used to control motor speed via the motor driver's enable pins, while direction is controlled through the input logic pins.

3.5 Power Distribution System

The entire system is powered from a single regulated 12V DC power supply rated at 5A. The power distribution architecture is as follows: the 12V rail feeds the L298N motor drivers and stepper motor drivers directly, since these components operate at 12V. The Arduino Nano and the robotic arm's NodeMCU are powered from the 5V regulated output of the Arduino CNC shield, which derives 5V from the 12V rail via an

onboard linear regulator. The mobile base's NodeMCU and joystick circuitry are powered by a 7805 voltage regulator circuit tapped from the 12V supply.

Decoupling capacitors (100µF electrolytic and 100nF ceramic) are placed across the power supply pins of each microcontroller and motor driver module to suppress voltage transients caused by motor switching. This is critical for preventing microcontroller resets and communication errors during operation.

CHAPTER 4: SOFTWARE IMPLEMENTATION

4.1 Software Architecture Overview

The software for this system is developed in the Arduino IDE using the C/C++ programming language and relevant libraries. The overall software architecture is divided into three layers: the Hardware Abstraction Layer (HAL), the Communication Layer, and the Application Layer. The HAL provides standardized interfaces to the GPIO, PWM, UART, and stepper driver hardware. The Communication Layer handles the ESP-NOW protocol initialization, peer registration, data packet serialization/deserialization, and transmission/reception callbacks. The Application Layer implements the joystick input processing, motor control logic, motion recording state machine, and playback execution engine.

4.2 ESP-NOW Communication Protocol Implementation

The ESP-NOW protocol is initialized on both the transmitter (controller) and receiver (robot) NodeMCU modules. On the transmitter side, the following initialization sequence is followed: the ESP8266/ESP32 Wi-Fi module is initialized in WIFI_STA (Station) mode, ESP-NOW is initialized using the esp_now_init() function, the receiver's MAC address is registered as a peer using esp_now_add_peer(), and a send callback function is registered to handle transmission acknowledgements.

The data packet structure transmitted over ESP-NOW is defined as a C struct to ensure a compact and well-defined binary layout:

Data Struct	*typedef struct { int joyX; int joyY; int joyRot; int armBase; int armShoulder; int armElbow; int armWrist1; int armWrist2; int gripper; bool recordBtn; bool playBtn; } ControlPacket;*

This structure occupies 40 bytes, well within the 250-byte payload limit of ESP-NOW. The joystick analog values are mapped from the ADC range (0–1023 on ESP8266 and 0–4095 on ESP32) to signed integer values in the range -255 to +255, where 0 represents the center (neutral) position. A dead-band of ±10 units is applied around the center to suppress noise from the joystick potentiometers.

On the receiver side, an esp_now_register_recv_cb() callback function is registered. Upon receipt of a packet, the callback copies the received data into a global volatile

ControlPacket buffer and sets a newDataFlag. The main loop polls this flag and processes the data accordingly. This design decouples the time-critical reception callback from the application processing logic.

4.3 Stepper Motor Control Algorithm

Stepper motor control is implemented using the AccelStepper library, which supports both constant-speed and acceleration-controlled stepping profiles. Each of the five arm joints is represented as an AccelStepper object initialized with the DRIVER interface mode (step and direction pins). The maximum speed and acceleration for each joint are configured based on the joint's mechanical reduction ratio and the A4988 driver's microstepping setting (set to 1/8 microstepping in this project).

Joystick commands are translated into stepper motor position increments. A command value of +100 (representing full deflection in one direction) is mapped to an increment of +50 steps per loop iteration, while -100 maps to -50 steps. The AccelStepper library's moveTo() and run() functions handle the trapezoidal velocity profile automatically, preventing sudden velocity changes that would cause stepper motor slip or vibration.

A software end-stop check is implemented for each joint, preventing commands that would drive a joint beyond its physical range of motion. If a position command exceeds the defined limit, it is clamped to the limit value. This prevents mechanical damage during operation.

4.4 Mecanum Wheel Drive Algorithm

The mecanum wheel drive algorithm runs on the receiver NodeMCU and computes individual wheel speed setpoints from the received joystick translational and rotational velocity commands. The kinematic equations implemented are derived from the 3-wheel mecanum model presented in Chapter 2:

Code	*int w1 = Vy + Vx + Wrot; int w2 = Vy - Vx - Wrot; int w3 = Vy - Wrot;*

Each computed wheel speed is clamped to the range -255 to +255 (corresponding to full reverse to full forward PWM duty cycle). The sign of each value determines the direction pin state of the corresponding L298N channel, while the absolute value is written to the PWM enable pin using analogWrite(). A minimum speed threshold of

PWM=30 is applied to overcome motor and gearbox static friction, below which the motor is stopped completely to prevent jitter.

4.5 Motion Recording and Playback System

The motion recording system captures and stores the complete robot state at regular time intervals. The recording is triggered by pressing the dedicated record button on the joystick controller, and stopped by pressing it again. During recording, a timestamped snapshot of the complete control state is appended to a dynamically allocated array in the NodeMCU's RAM at a rate of 20 Hz (50 ms intervals).

The recording data structure stores the complete ControlPacket along with a timestamp:

Struct	*struct RecordFrame { uint32_t timestamp_ms; ControlPacket data; };*

The maximum recording duration is limited by available RAM. On the ESP8266, with 80 KB of usable heap, approximately 1,600 frames can be stored, corresponding to 80 seconds of recording at 20 Hz. On the ESP32 with 320 KB heap, this increases to approximately 6,400 frames (320 seconds). For longer recordings, an external SD card module can be added via SPI, a capability that is identified as a future enhancement.

During playback, the stored frames are replayed in sequence. The playback engine reads each frame, transmits the stored ControlPacket data to the robot (simulating real-time joystick input), waits for the interval duration (timestamp of next frame minus current frame), and then advances to the next frame. This timestamp-based replay ensures that motion timing is accurately reproduced regardless of the execution speed of the loop.

4.6 Serial Communication between NodeMCU and Arduino

The receiver NodeMCU communicates with the Arduino Nano via a hardware UART link at 115200 baud. The arm control data (5 joint values and gripper) is serialized into a comma-separated string terminated by a newline character, for example: '120,45,-30,0,90,45\n'. This format is simple, human-readable for debugging, and reliably parsed by the Arduino's Serial.parseInt() functions. The NodeMCU transmits an updated string only when the arm joint values have changed by more than a threshold of 5 steps, reducing unnecessary serial traffic.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Experimental Setup

The complete robotic system was assembled and tested in the Electronics and Communication Engineering laboratory at Central University of Jammu. Functional testing was conducted across five performance dimensions: communication latency, drive accuracy, arm positioning accuracy, motion playback fidelity, and power consumption. All tests were conducted over a minimum of 30 trials to ensure statistical reliability of the results.

5.2 Communication Performance

The ESP-NOW communication latency was measured by recording the timestamp of packet transmission and comparing it with the timestamp of packet reception, using synchronized system clocks on the transmitter and receiver NodeMCU modules. Tests were conducted at distances ranging from 1 meter to 50 meters, in line-of-sight conditions.

Distance (m)	Min Latency (ms)	Avg Latency (ms)	Max Latency (ms)	Packet Loss (%)
1	4.2	5.8	9.1	0.0%
5	4.5	6.3	11.4	0.0%
10	5.1	7.2	13.8	0.0%
20	6.3	9.4	17.6	0.2%
30	7.8	12.1	21.3	0.5%
50	9.2	16.8	28.5	1.8%

The results confirm that ESP-NOW provides excellent low-latency performance at distances up to 30 meters, with average latency comfortably below 20 ms. At 50 meters, the packet loss rate of 1.8% is acceptable for a non-critical control application but may warrant the implementation of a retransmission strategy for safety-critical deployments. These results are consistent with published ESP-NOW performance evaluations in the literature.

5.3 Arm Positioning Accuracy

Arm positioning accuracy was evaluated by commanding each joint to a series of target positions (0°, 30°, 60°, 90°, 120° where applicable) and measuring the actual achieved angle using a digital protractor. Five trials were conducted per target position per joint, and the mean absolute error (MAE) was calculated.

Joint	Target Range	Mean Abs. Error	Max Error	Std Dev
Base (J1)	0°–360°	±1.2°	±2.5°	0.6°
Shoulder (J2)	-30°–+90°	±0.9°	±1.8°	0.4°
Elbow (J3)	0°–135°	±1.1°	±2.1°	0.5°
Wrist Pitch (J4)	-90°–+90°	±1.5°	±3.0°	0.8°
Wrist Roll (J5)	-180°–+180°	±1.8°	±3.5°	0.9°

The observed positional errors are primarily attributable to belt elasticity and backlash in the pulley transmission system, mechanical play in the 3D-printed joint interfaces, and minor thermal expansion of the PLA structural components during extended operation. These errors are acceptable for the intended educational and research applications of the system and are comparable to those reported for similar low-cost robotic arm implementations in the literature.

5.4 Omnidirectional Motion Performance

The mecanum wheel base was evaluated by commanding specific directional motions (forward, lateral, diagonal, and rotation) and measuring the actual trajectory using a tape measure and reference markers on the floor. Trajectory accuracy was defined as the deviation from the commanded straight-line path over a 1-meter distance.

Forward and backward motion showed a mean path deviation of 12 mm over 1 meter (1.2%), which is acceptable for a low-cost mecanum implementation. Lateral (sideways) motion showed a larger deviation of 28 mm over 1 meter (2.8%), attributable to uneven roller contact and slight misalignment of the wheel mounting brackets. Rotational motion was accurate to within ±2.5° per 360° revolution. Overall, the base demonstrated reliable omnidirectional capability for its intended use cases.

5.5 Motion Playback Accuracy

Motion playback accuracy was evaluated by recording a standardized 10-step pick-and-place sequence and replaying it 30 times. The end-effector position at each of the 10 key waypoints was measured in each replay trial, and the deviation from the positions achieved during the original recording was computed.

Across 30 trials and 300 total waypoint measurements, 94.7% of end-effector positions were within 5 mm of the recorded reference position. The remaining 5.3% of cases showed deviations of 5–12 mm, concentrated at the wrist joints (J4, J5), which have slightly higher positional errors as noted in Section 5.3. No playback failures (complete misses or collisions) were observed across all 30 trials. This result validates the motion recording system as reliable for repetitive task automation.

5.6 Comparison with Existing Systems

Parameter	This Project	Similar System A*	Similar System B*
DOF	5+Gripper	4+Gripper	6+Gripper
Mobility	Omnidirectional	Differential Drive	Fixed
Wireless Protocol	ESP-NOW	Wi-Fi UDP	RF 433 MHz
Avg. Latency	~8 ms	~35 ms	~15 ms
Motion Recording	Yes	No	Yes
Approx. Cost	~Rs. 5,000	~Rs. 8,500	~Rs. 12,000
Playback Accuracy	~95%	N/A	~88%

*Similar System A and B refer to representative low-cost robotic arm and mobile base systems described in published literature. Specific citations are provided in the References section.

5.7 Limitations and Observations

Several limitations were identified during experimental evaluation. First, the 3-wheel mecanum configuration provides less stability than a 4-wheel design when the arm is fully extended and carrying a payload, due to the reduced support polygon. Second, the arm's PLA structural components exhibit slight deformation under sustained load, particularly at the shoulder and elbow joints. Third, the absence of position feedback encoders on the stepper motors means that stalling (caused by sudden load changes

or impacts) results in permanent positional error accumulation. Fourth, the maximum recording duration is limited by available MCU RAM, restricting recordings to approximately 80–320 seconds depending on the MCU used.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project has successfully designed, implemented, and validated a mobile robotic system that integrates a BCN3D Moveo open-source 5 DOF robotic arm with a 3-wheel mecanum omnidirectional mobile base, controlled wirelessly via the ESP-NOW protocol with motion recording and playback capability. The system has been realized at an approximate cost of Rs. 5,000, demonstrating the feasibility of building a functional and versatile robotic platform within a stringent budget.

The key contributions of this project are: the successful mechanical integration of a 5 DOF arm on a mobile base; the implementation of a low-latency, infrastructure-free ESP-NOW wireless control system with average latency below 10 ms at typical operating distances; the development of a timestamp-based motion recording and playback system achieving approximately 95% waypoint accuracy; and the derivation and implementation of a 3-wheel mecanum kinematic model enabling omnidirectional mobility.

The project demonstrates strong command of embedded systems design, motor control theory, wireless communication protocols, and robotics kinematics. It provides a solid, extensible platform for further research and development in educational robotics. The detailed documentation provided in this report enables full reproducibility of the system by other researchers and students.

6.2 Future Scope

The following directions are identified for future development of this system:

- ROS Integration: Porting the control system to the Robot Operating System (ROS2) would enable the use of standardized navigation, manipulation, and perception packages, significantly expanding the system's capabilities.
- Computer Vision: Adding a Raspberry Pi Camera Module and implementing OpenCV-based object detection would enable the system to autonomously locate, grasp, and place objects, transforming it from a teleoperated system to a semi-autonomous one.
- Closed-Loop Motor Control: Adding rotary encoders or Hall-effect sensors to each stepper motor joint would enable closed-loop position control, eliminating positional drift caused by motor stalling and improving accuracy.

- Battery Power: Replacing the bench power supply with a lithium-polymer battery pack and implementing a battery management system (BMS) would make the robot fully mobile and untethered.
- Inverse Kinematics: Implementing an IK solver (e.g., using the IKFAST plugin or a custom DH-parameter-based solution) would allow the operator to specify desired end-effector positions in Cartesian space rather than individual joint angles.
- Extended Recording Storage: Adding an SD card module via SPI would enable unlimited recording duration, removing the RAM constraint on the current motion recording system.
- Multi-Robot Coordination: Leveraging ESP-NOW's support for multiple peers, a future implementation could coordinate multiple robots to collaborate on shared tasks.

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APPENDIX A: CIRCUIT DIAGRAMS AND PIN MAPS

A.1 Arduino Nano CNC Shield Pin Mapping

CNC Shield Pin	A4988 Driver	Stepper Motor Joint	Step Pin (Nano)	Dir Pin (Nano)
X-Axis	Driver 1	Base (J1)	D2	D5
Y-Axis	Driver 2	Shoulder (J2)	D3	D6
Z-Axis	Driver 3	Elbow (J3)	D4	D7
A-Axis	Driver 4	Wrist Pitch (J4)	D12	D13
(External)	Driver 5	Wrist Roll (J5)	A0	A1
Servo Pin	MG996R Servo	Gripper	D11	N/A

A.2 NodeMCU (Receiver) Pin Mapping for L298N Motor Drivers

NodeMCU GPIO	L298N Module	L298N Pin	Function	Wheel
D1 (GPIO5)	Module 1	IN1	Direction A	Wheel 1
D2 (GPIO4)	Module 1	IN2	Direction B	Wheel 1
D3 (GPIO0)	Module 1	ENA (PWM)	Speed	Wheel 1
D5 (GPIO14)	Module 1	IN3	Direction A	Wheel 2
D6 (GPIO12)	Module 1	IN4	Direction B	Wheel 2
D7 (GPIO13)	Module 1	ENB (PWM)	Speed	Wheel 2
D8 (GPIO15)	Module 2	IN1	Direction A	Wheel 3
D0 (GPIO16)	Module 2	IN2	Direction B	Wheel 3
RX (GPIO3)	Module 2	ENA (PWM)	Speed	Wheel 3

A.3 ESP-NOW MAC Address Configuration

For the ESP-NOW system to function correctly, the receiver NodeMCU's MAC address must be hard-coded into the transmitter firmware. The MAC address can be read from the receiver module using the following Arduino sketch snippet:

Code	*#include <ESP8266WiFi.h> void setup() { Serial.begin(115200); Serial.println(WiFi.macAddress()); } void loop() {}*

The resulting 6-byte MAC address (e.g., AA:BB:CC:DD:EE:FF) must then be entered as the 'broadcastAddress' array in the transmitter firmware. This step is required once during initial system configuration and need not be repeated unless the receiver module is replaced.